

	graph at a lower maximum speed.
7(a)	The origin of upthrust is due to the difference in fluid pressure between the lower surface and upper surface of a body.
7(b)	Since the tube floats in the liquid,

	Weight of tube and sand = upthrust exerted by liquid on tube and sand $mg = V\rho g$ $mg = Ah\rho g$ $m = Ah\rho$
7(c)(i)	In the expression, ρ, A, g and m are constants. The acceleration a is therefore proportional to the displacement x from the equilibrium position. The negative sign in the expression shows that the direction of the acceleration a is always in the opposite direction to the displacement x from the equilibrium position. These conditions for the tube to perform simple harmonic motion is thus satisfied.
7(c)(ii)	Compare the expression $a = -\left(\frac{\rho Ag}{m}\right)x$ with $a = -\omega^2 x$, $\omega^2 = \left(\frac{\rho Ag}{m}\right)$ $(2\pi f)^2 = \left(\frac{\rho Ag}{m}\right)$ $f = \sqrt{\frac{\rho Ag}{4\pi^2 m}}$ $f = \sqrt{\frac{(1.0 \times 10^3)(4.2 \times 10^{-4})9.81}{4\pi^2 (32 \times 10^{-3})}}$ $f = 1.8 \text{ Hz}$
7(d)(i)1.	$f = \frac{1}{T}$ $= \frac{1}{0.50}$ $= 2.0 \text{ Hz}$
7(d)(i)2.	$f = \sqrt{\frac{\rho Ag}{4\pi^2 m}}$ $2.0 = \sqrt{\frac{\rho(4.2 \times 10^{-4})9.81}{4\pi^2 (32 \times 10^{-3})}}$ $\rho = 1.2 \times 10^3 \text{ kg m}^{-3}$
7(d)(ii)1.	1. The tube loses energy due to the viscous forces of the liquid. This causes the oscillations to be damped. 2. The tube loses energy to the surroundings and there is no work done by an external force on the tube to compensate for the energy lost.

7(d)(ii)2.	When $t = 0.0 \text{ s}$, $x = 1.50 \text{ cm}$, When $t = 1.0 \text{ s}$, $x = 0.85 \text{ cm}$. Total energy $= \frac{1}{2} m \omega^2 x_0^2 = \frac{1}{2} m (2\pi f)^2 x_0^2$ Decrease in energy $= \frac{1}{2} (32 \times 10^{-3}) 4\pi^2 (2.0)^2 [(1.50 \times 10^{-2})^2 - (0.85 \times 10^{-2})^2]$ $= 3.9 \times 10^{-4} \text{ J}$
8(a)(i)	